INFOSYS SPRINGBOARD

Receipt & Invoice Digitizer

AI-powered receipt analysis, OCR extraction & expense management

Team D

Final Presentation · 2026

Meet the Team

Team D · Infosys Springboard

Sriya Muthukumar

Frontend

@Sriyamk

Landing page, bill digitization page, receipt content mapping

Shaik Mastan

Database & Admin

@skmdmastan07

Database design, Admin dashboard, admin user receipts page

Trishala Koduru

Research & UX

@Thrishala2006

OCR research, forgot password, registration, multilingual & dark mode

Atul Chandra

AI & Backend

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OCR integration, chatbot (Mistral/Ollama), backend logic

Rithwik Pachipala

Auth & Security

@Rithwik-18

Login page, reset password page, session management

Shailee Chouksey

Documentation

Project documentation, reports and presentation materials

Problem Statement

What problem are we solving?

📄 Manual Receipt Management

Businesses and individuals manually sort, store and track paper receipts — a slow, error-prone process.

🔍 No Searchable Records

Physical receipts can't be searched or filtered by vendor, date or amount, making tax filing and audits difficult.

💸 Poor Expense Visibility

Without a central dashboard, tracking monthly spending patterns, top vendors and tax totals is impossible.

🤖 No AI Assistance

Users have no intelligent assistant to query their own spending data in plain natural language.

Project Objectives

01 Digitize physical receipts and invoices using AI-powered OCR (PaddleOCR + Mistral)

02 Automatically extract key fields: Vendor, Invoice Number, Date, Total Amount & Tax

03 Store all receipt data in a structured SQLite database per user

04 Provide a spending analytics dashboard with charts and summary statistics

05 Enable an AI chatbot to answer natural language queries about expenses

06 Support multilingual interface, dark mode, admin panel and secure authentication

Architecture & Workflow

1. Upload
User uploads image/PDF via drag-drop or file picker

2. OCR
PaddleOCR extracts raw text from the receipt/invoice

3. Mistral LLM
Mistral parses OCR text → extracts structured fields

4. Database
Extracted data stored in SQLite per user account

5. Dashboard
User views receipts, analytics charts & spending stats

6. AI Chatbot
Mistral chatbot answers queries about user's expenses

System Architecture

Browser (HTML/CSS/JS) — Flask Backend — SQLite Database — PaddleOCR + Mistral — Ollama Server

Technology Stack

Frontend

- HTML5, CSS3, JavaScript (Vanilla)
- Chart.js — analytics charts
- i18next — multilingual support
- Space Grotesk + DM Sans fonts

Backend

- Python 3.11
- Flask 3.x — web framework
- Werkzeug — file handling & security
- bcrypt — password hashing
- PyJWT — Google OAuth tokens

AI & OCR

- PaddleOCR — text extraction
- PaddlePaddle — OCR backend
- Mistral (via Ollama) — LLM
- PyMuPDF (fitz) — PDF support
- OpenCV + NumPy — image processing

Database & Infra

- SQLite — lightweight relational DB
- Ollama — local LLM server
- python-dotenv — env management
- Flask-Mail — email support

Key Features

📤 Receipt Upload

Drag-drop or click to upload images and PDFs with real-time progress indicator

🔍 AI OCR Extraction

PaddleOCR + Mistral LLM automatically extracts vendor, date, invoice # and amounts

📊 Analytics Dashboard

Interactive Chart.js graphs showing spending over time with 4 summary stat pills

🤖 AI Chatbot

Mistral-powered chat widget answers natural language expense queries in real time

🌙 Dark Mode

Full dark/light theme toggle with CSS variables, persisted across sessions

🌐 Multilingual Support

i18next integration supporting English, Hindi, Telugu, Tamil, Kannada & Malayalam

🔐 Secure Auth

bcrypt password hashing, Google OAuth, session management, forgot/reset password

👑 Admin Dashboard

Admin view of all users, upload stats, activity overview and per-user receipt history

Feature Enhancements

What we added beyond the base

🌙 Dark Mode

NEW

Full dark/light theme toggle using CSS variables, persisted in localStorage

🤖 AI Chatbot Widget

NEW

Floating Mistral chatbot for natural language queries on spending data

📊 Analytics Charts

NEW

Chart.js spending-over-time graphs + 4 real-time summary stat pills

🔍 OCR Accuracy

IMP

PaddleOCR 3.x + Mistral for improved vendor, tax & total extraction

🌐 Multilingual UI

NEW

i18next integration — English, Hindi, Telugu, Tamil, Kannada, Malayalam

👑 Admin Dashboard

NEW

Full admin panel: user overview, upload stats, per-user receipt history

🔐 Forgot/Reset Password

NEW

Complete forgot password + email-based reset password flow

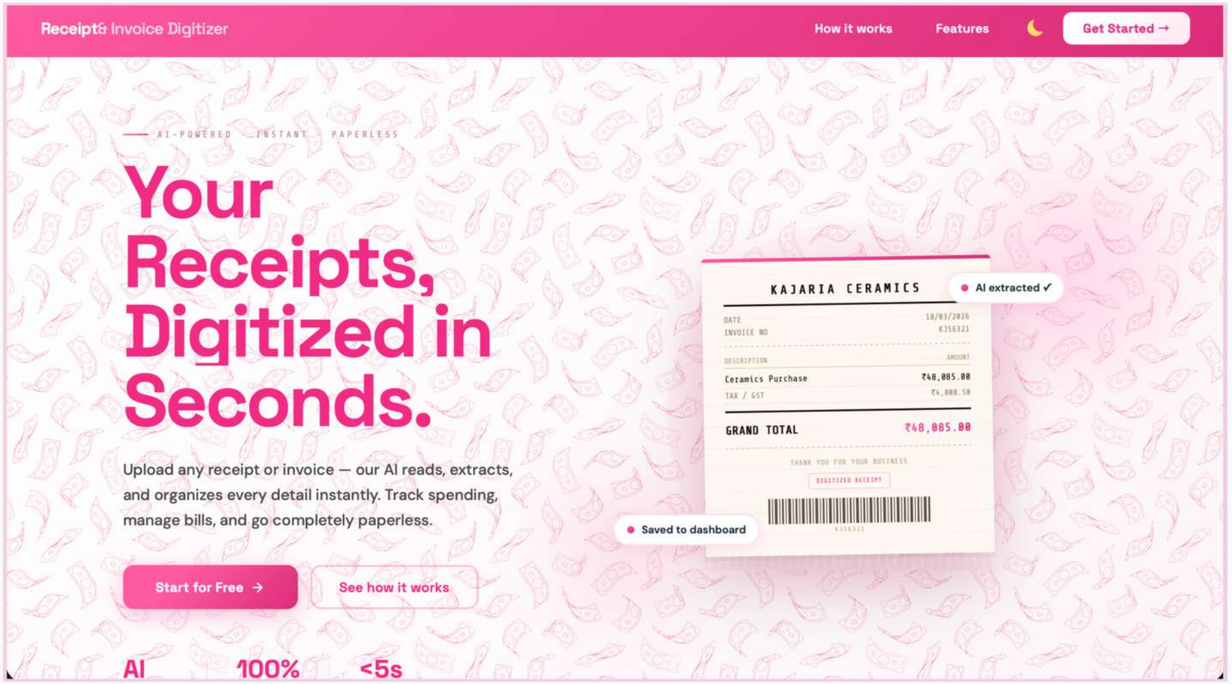
🎨 UI/UX Polish

IMP

Drag-drop upload, sticky navbar, upload progress spinner, responsive design

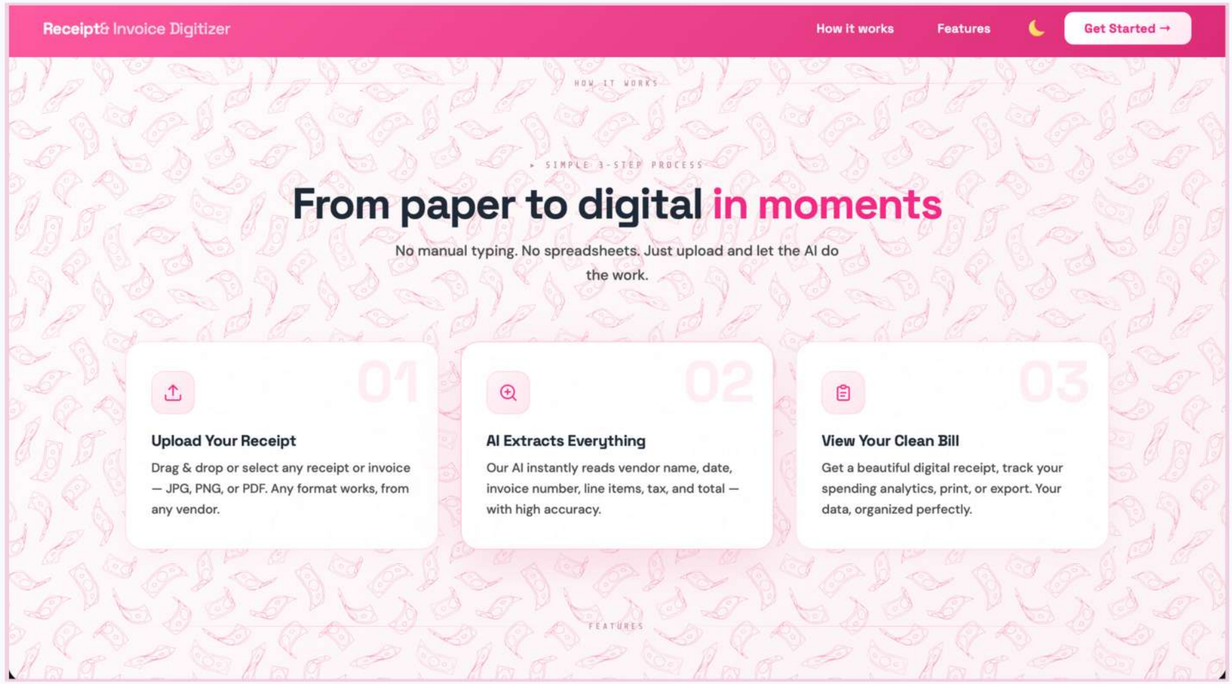
Application Screenshots

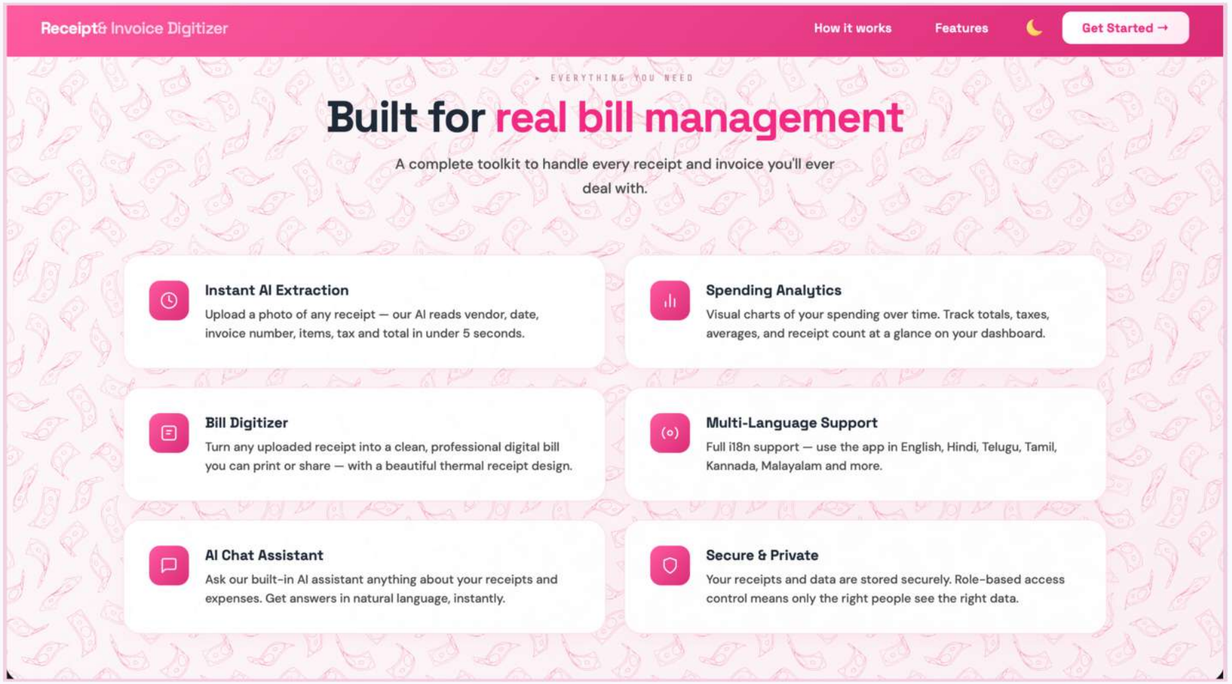
Video introduction and Landing Page



Application Screenshots

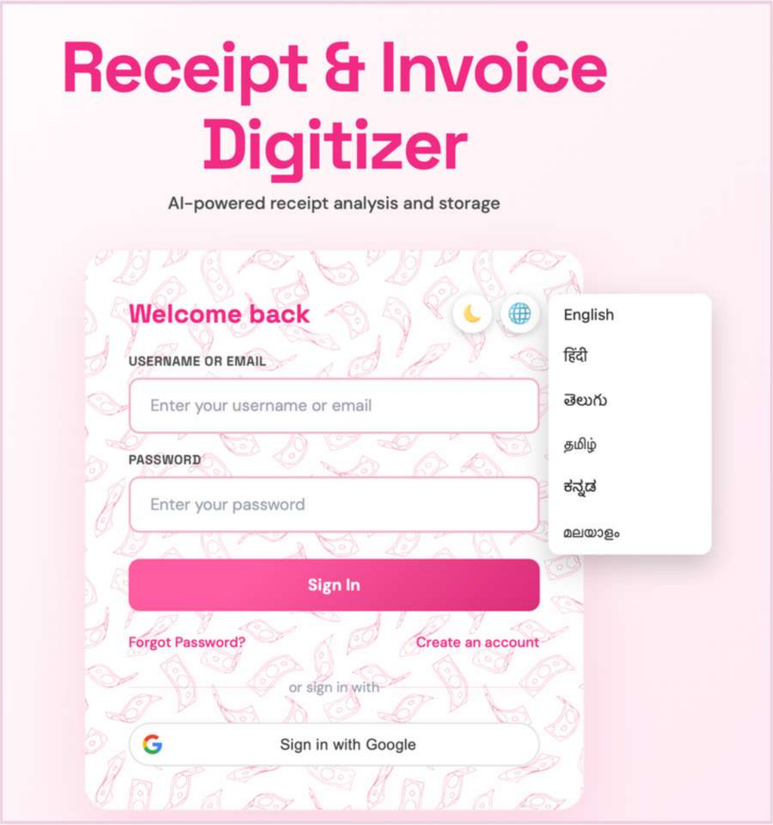
Landing Page

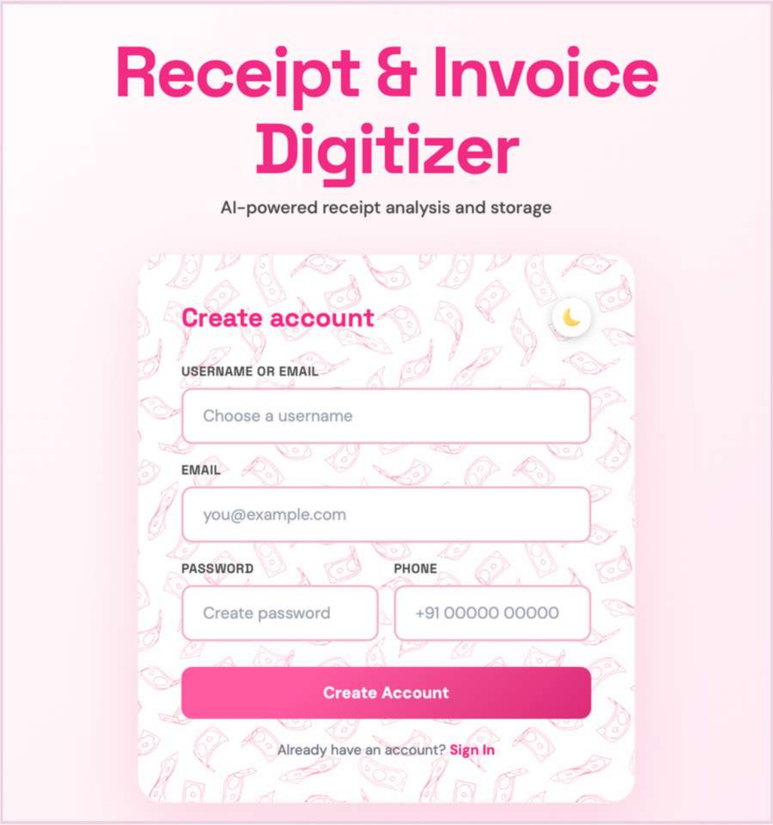


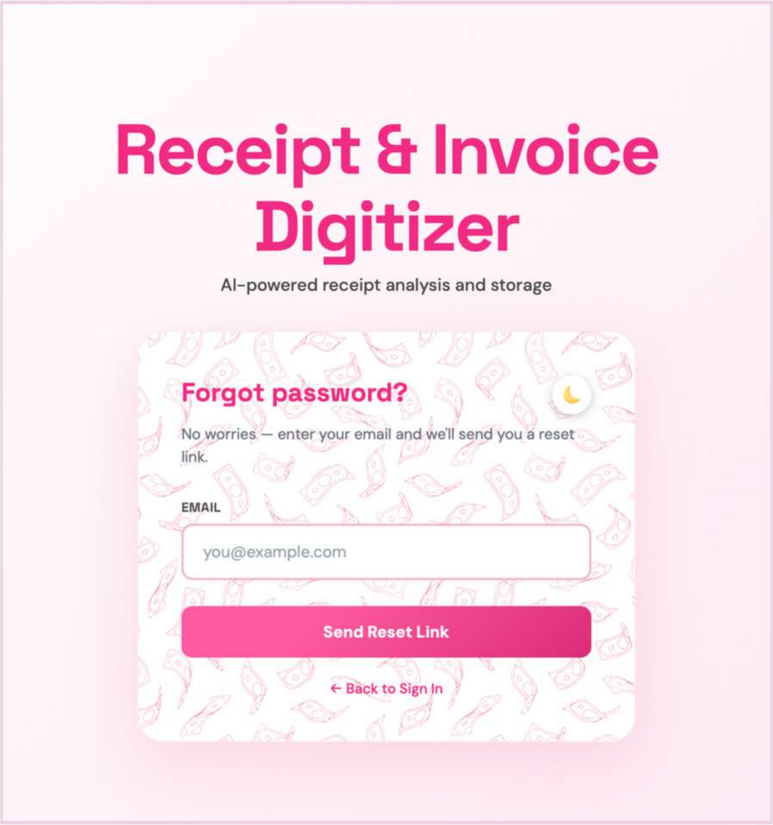


Application Screenshots

Login, signup, Forgot password pages

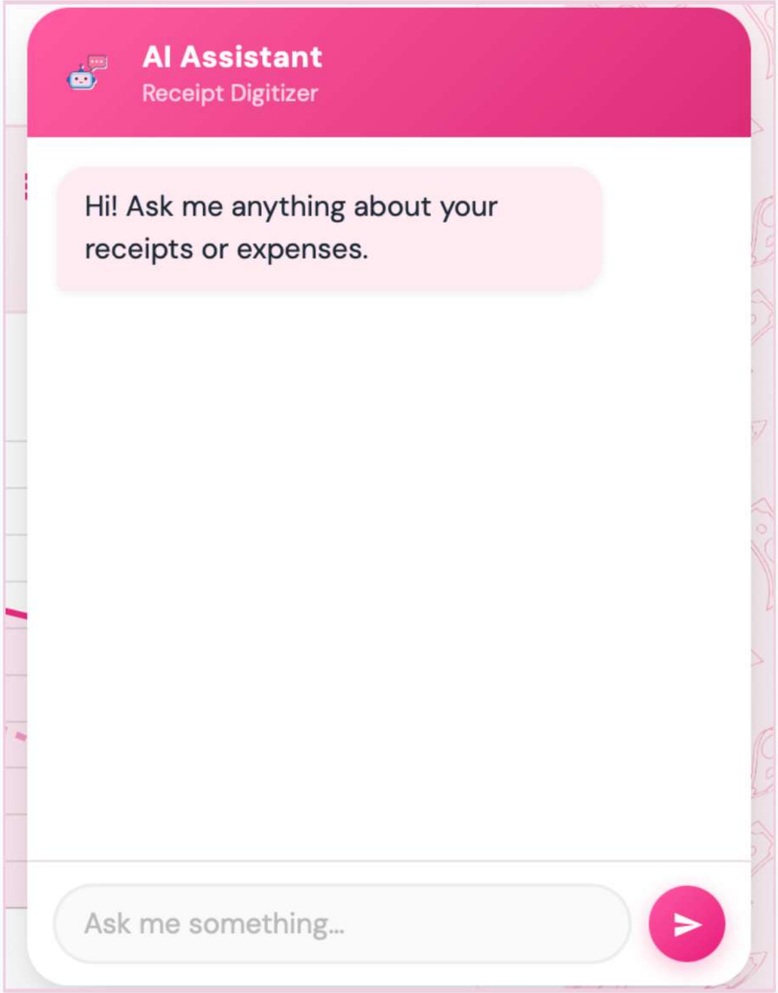


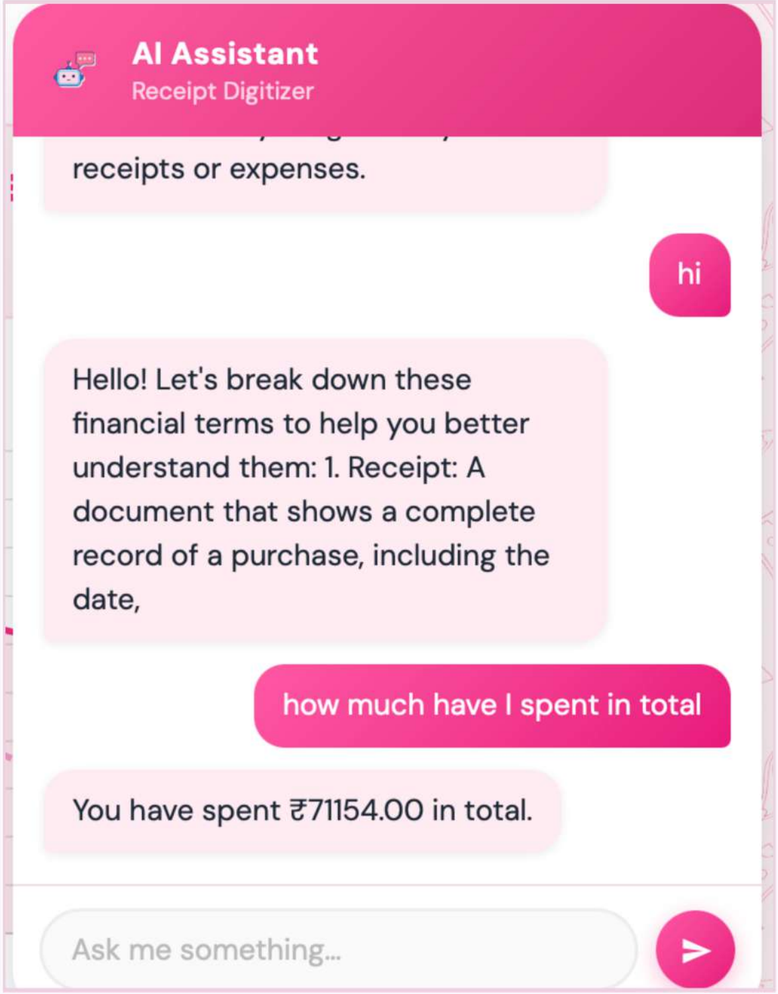


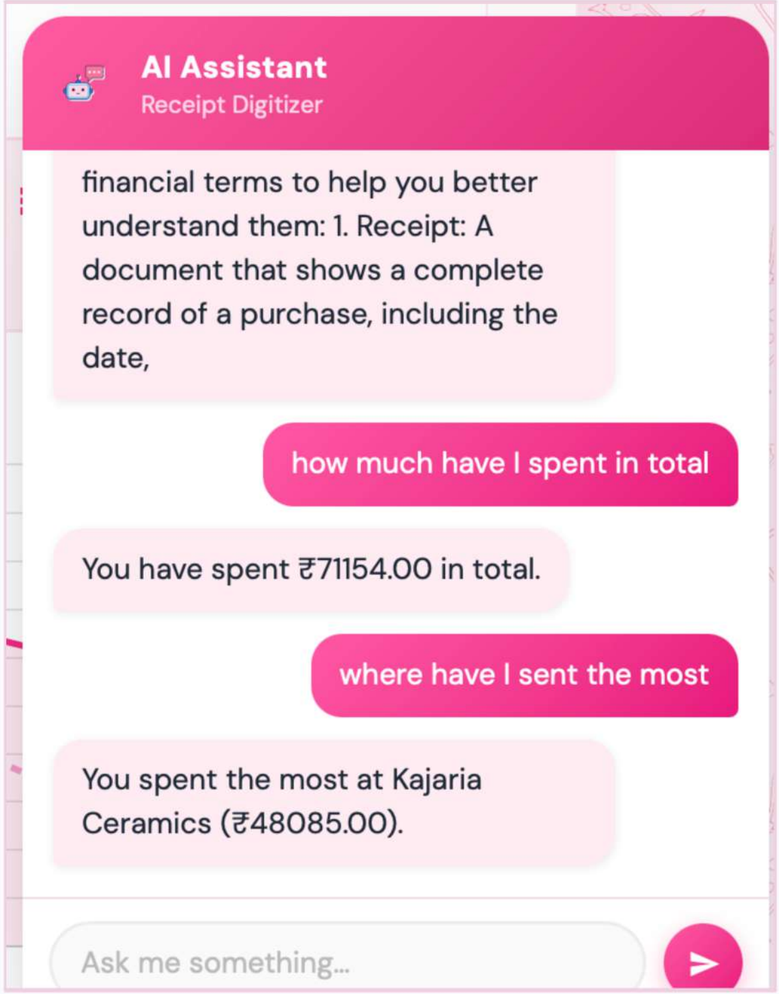


Application Screenshots

AI Assistant

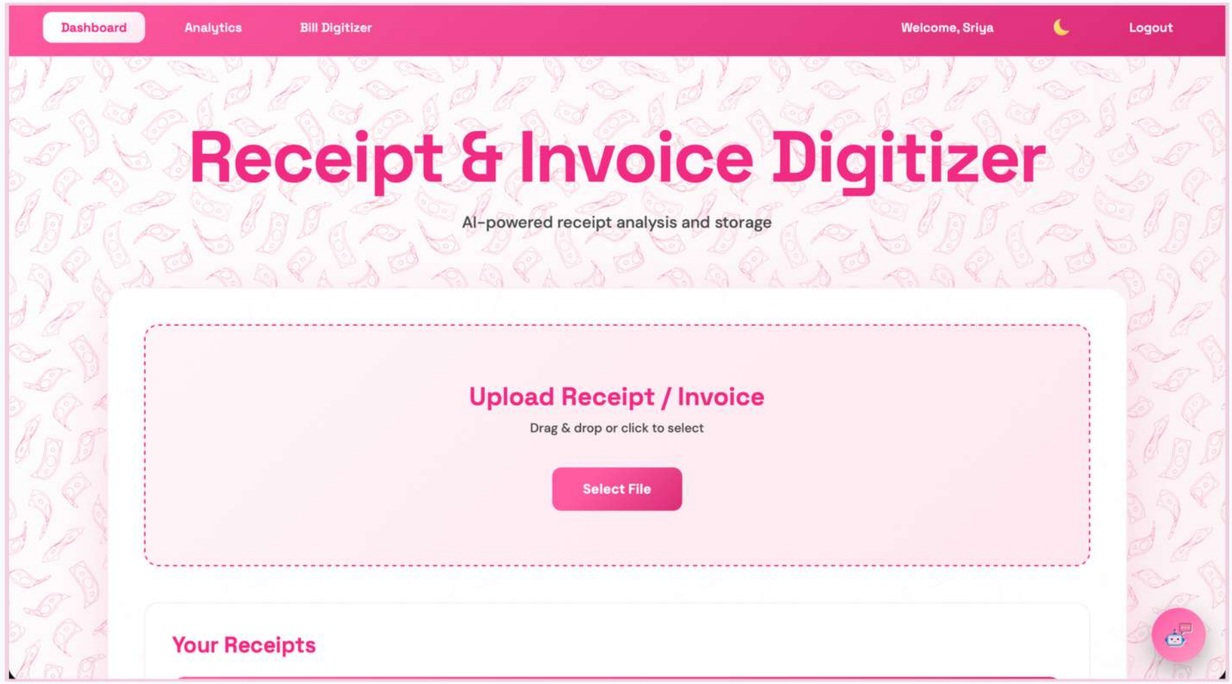


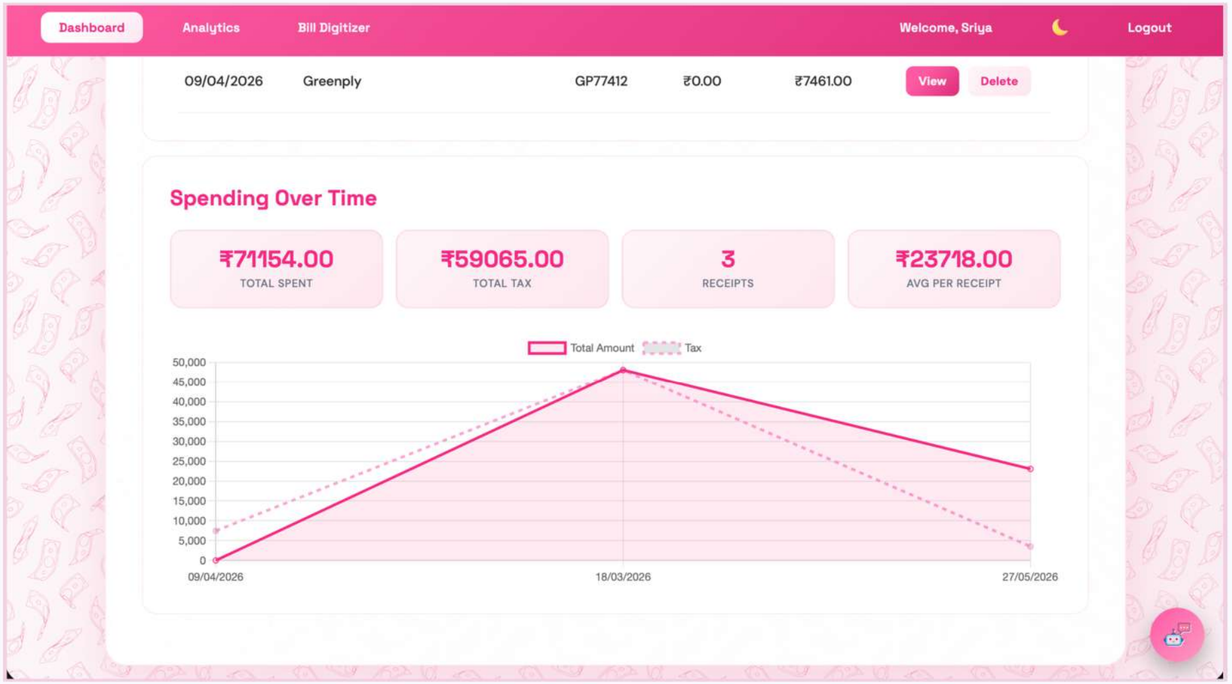




Application Screenshots

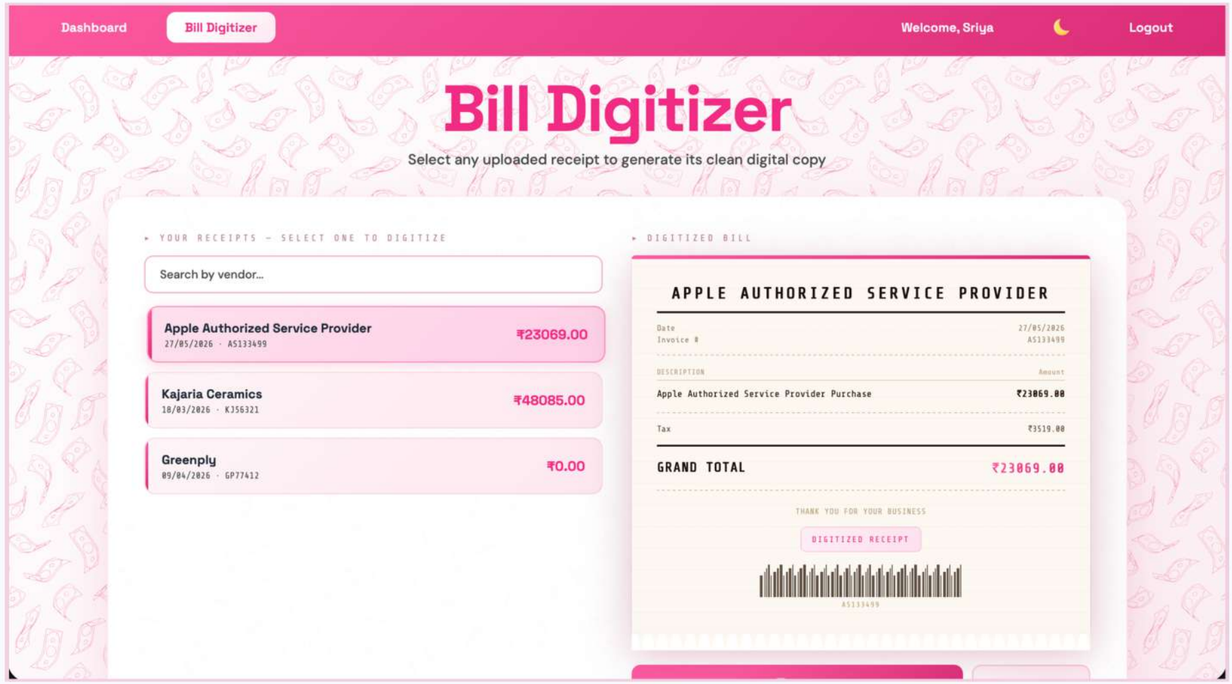
User Dashboard and Analytics

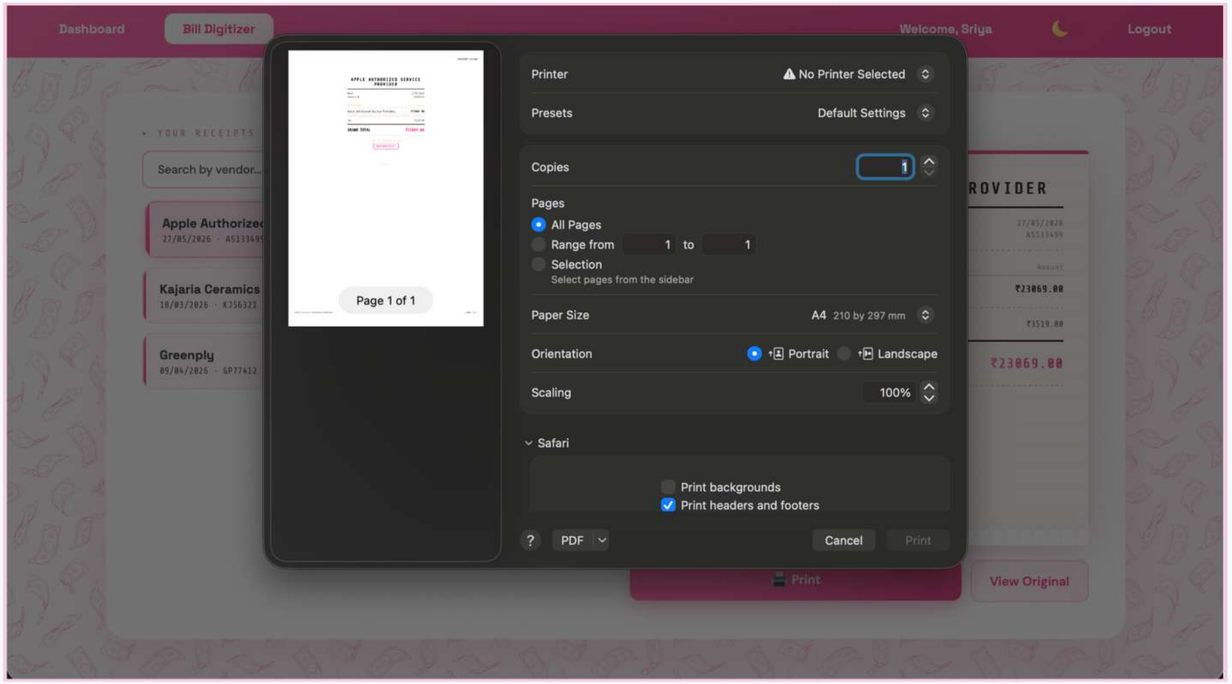




Application Screenshots

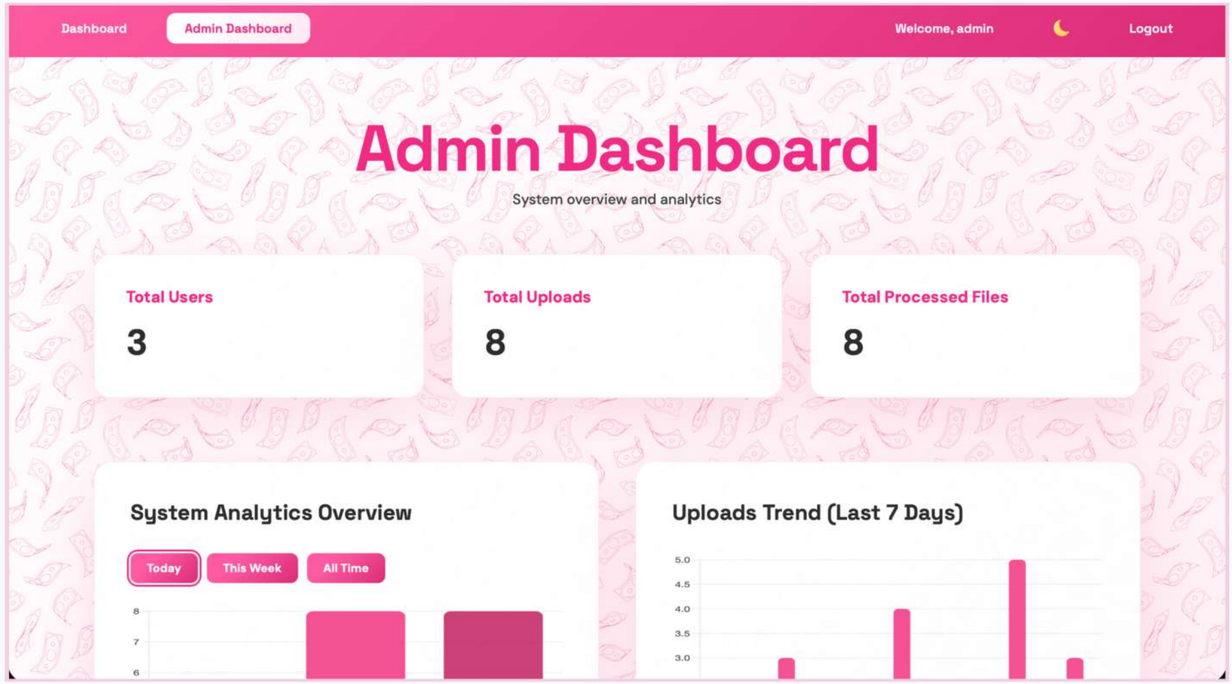
Bill Digitizer page and print option

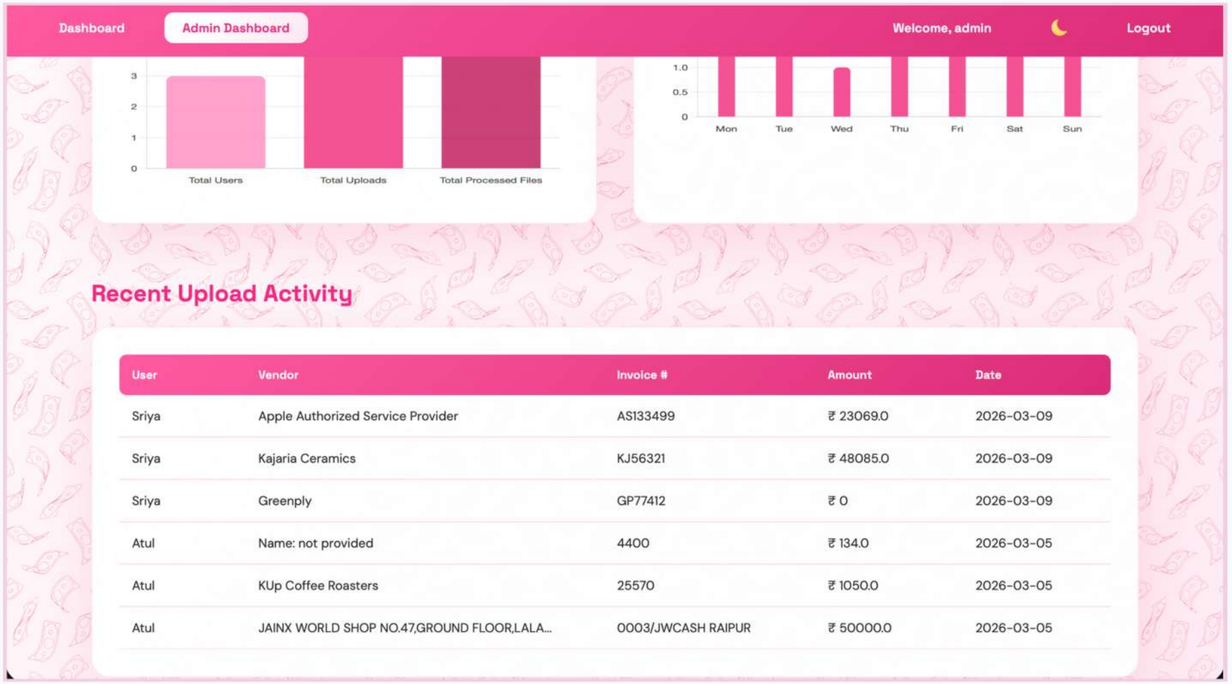




Application Screenshots

Admin Dashboard

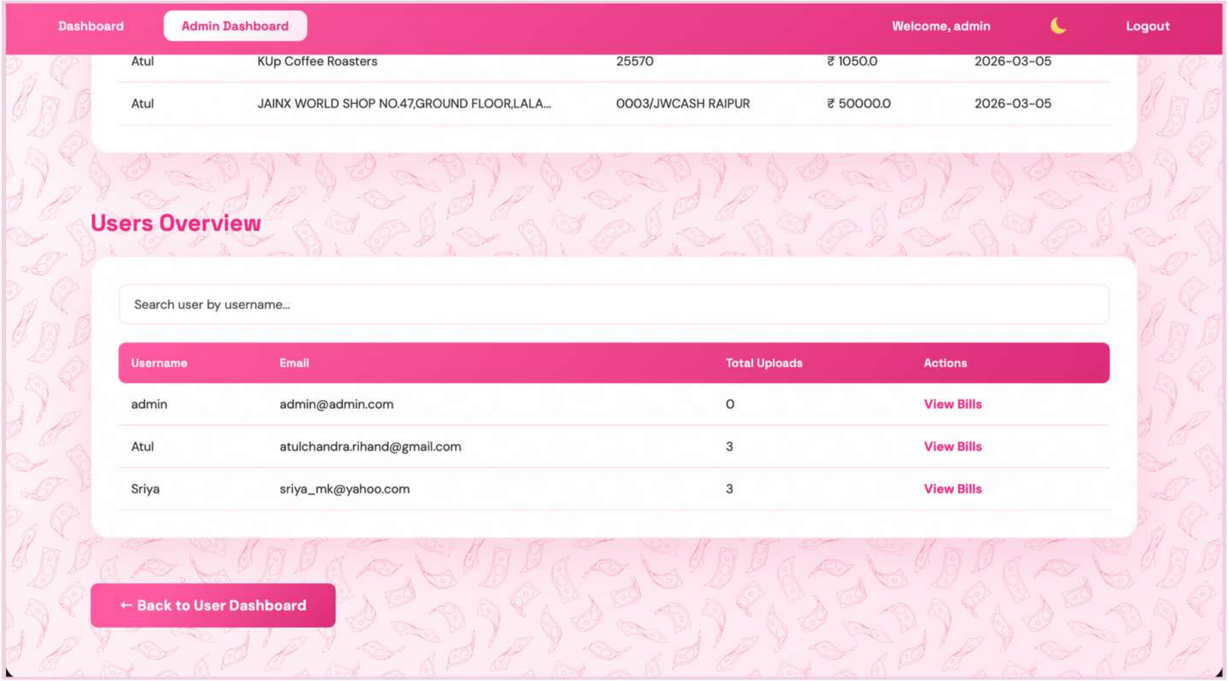


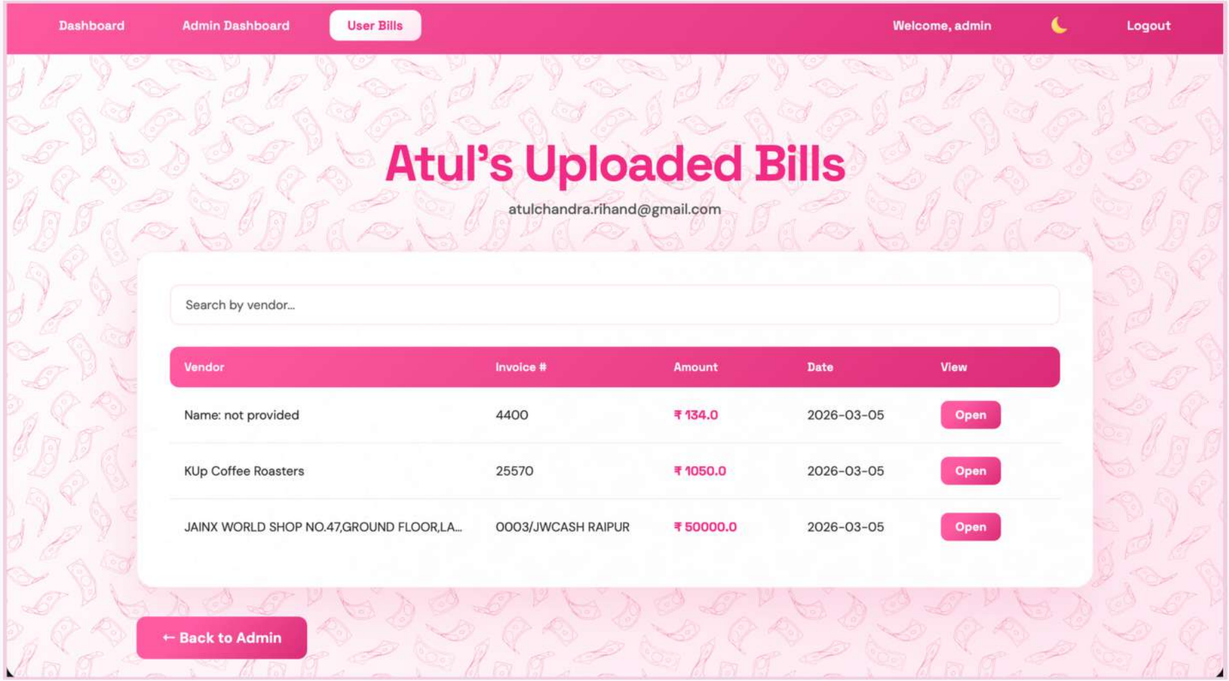


User	Vendor	Invoice #	Amount	Date
Sriya	Apple Authorized Service Provider	AS133499	₹ 23069.0	2026-03-09
Sriya	Kajaria Ceramics	KJ56321	₹ 48085.0	2026-03-09
Sriya	Greenply	GP77412	₹ 0	2026-03-09
Atul	Name: not provided	4400	₹ 134.0	2026-03-05
Atul	KUp Coffee Roasters	25570	₹ 1050.0	2026-03-05
Atul	JAINX WORLD SHOP NO.47,GROUND FLOOR,LALA...	0003/JWCASH RAIPUR	₹ 50000.0	2026-03-05

Application Screenshots

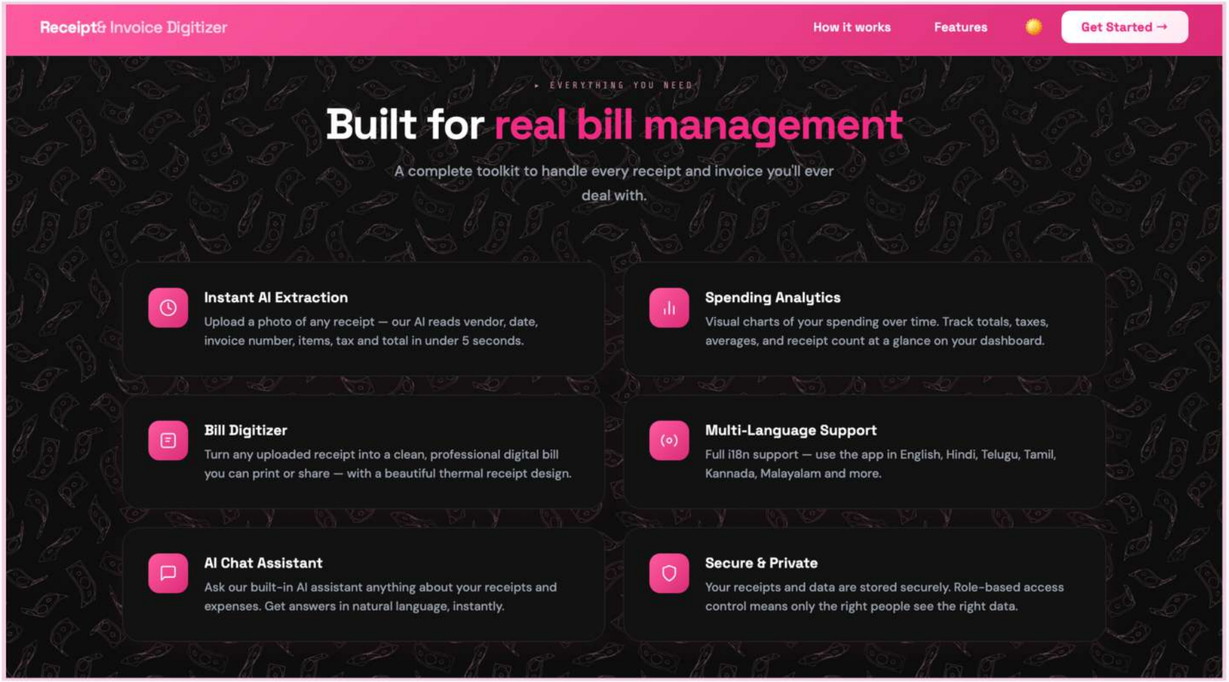
Admin Dashboard and User Bills page

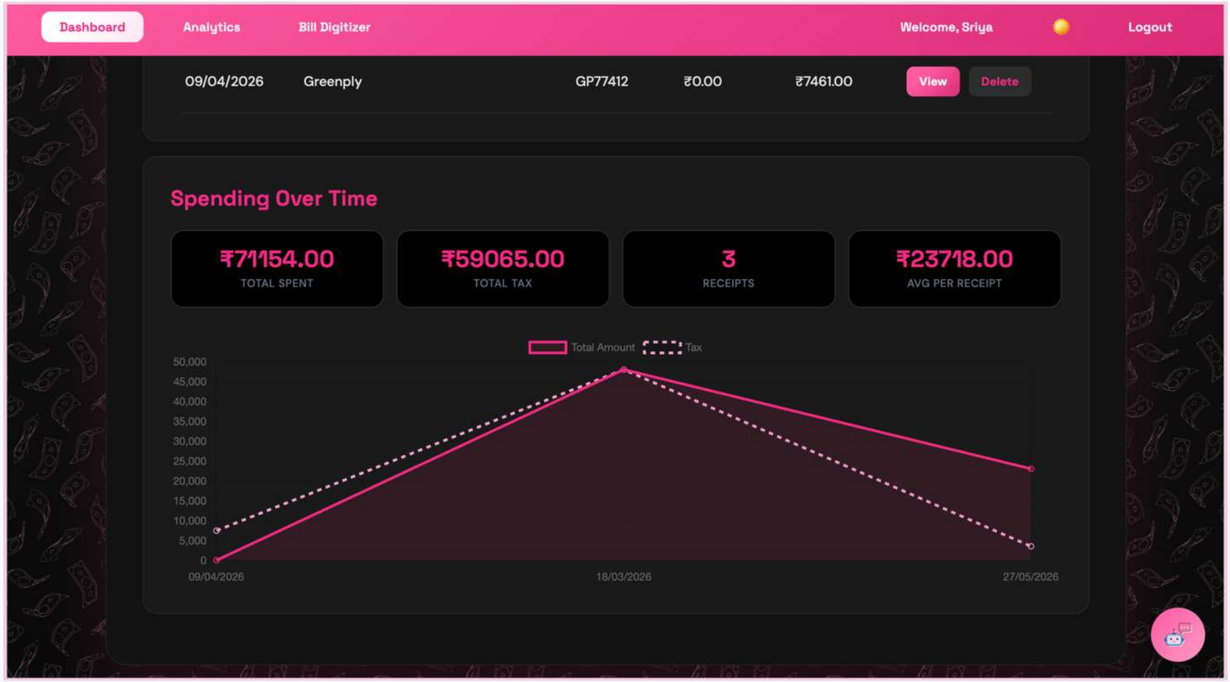




Application Screenshots

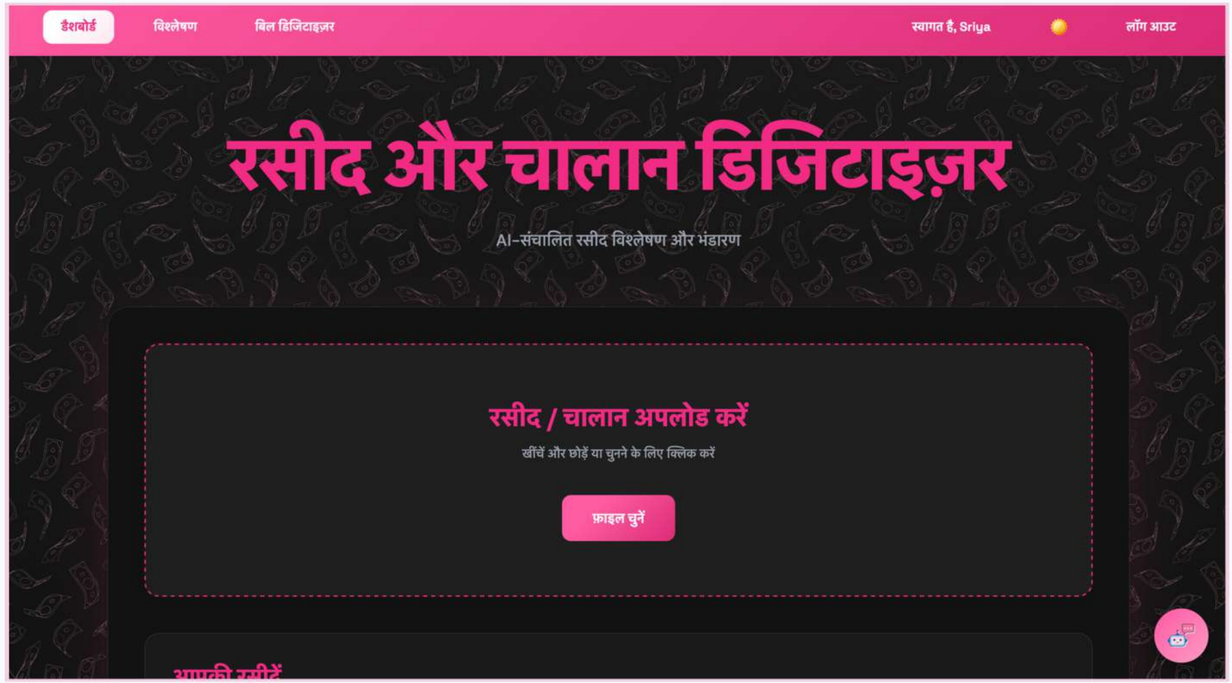
Dark mode

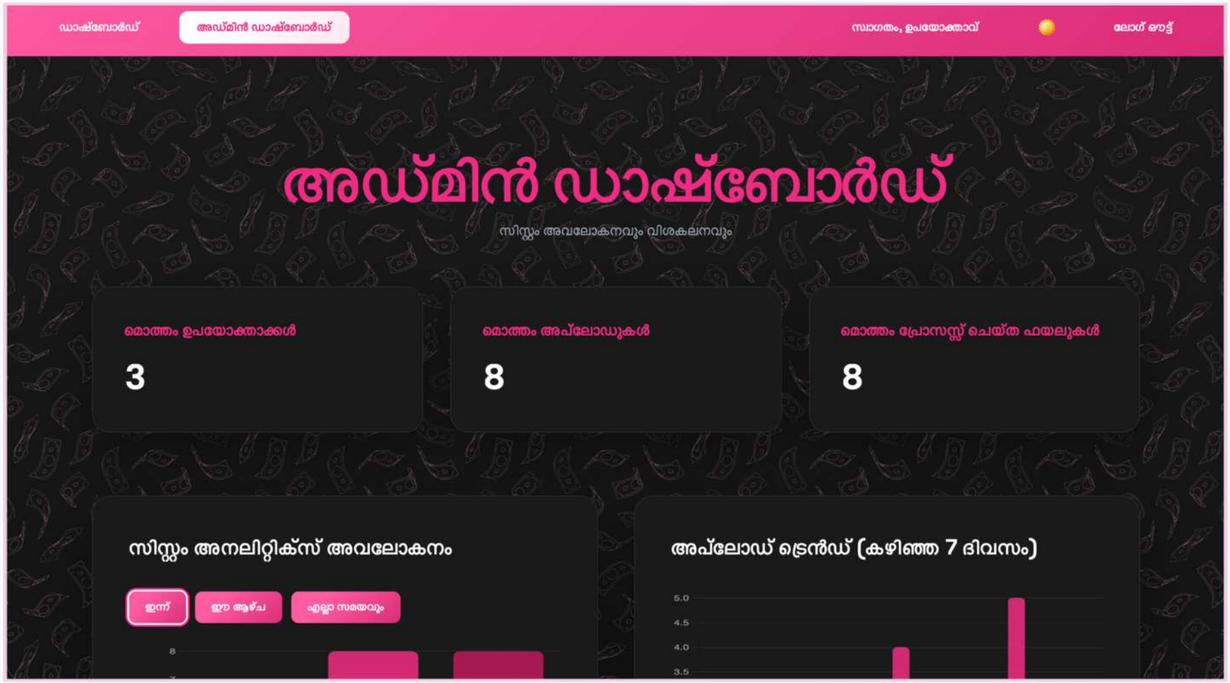




Application Screenshots

Multi Language opt





Challenges & Learnings

⚠️ Challenges

- PaddleOCR version differences (2.x vs 3.x) required custom result parsing
- Mistral hallucinating tax values on split-line invoice formats
- Ollama ConnectionError crashing Flask when server not running
- Git merge conflicts from parallel branch development by 6 members
- i18n welcome username getting overwritten by translation logic

💡 Learnings

- Integrating local LLMs (Mistral) for structured data extraction from unstructured text
- Building full-stack apps with Flask, SQLite and session-based authentication
- Collaborative Git workflows: branching, pull requests and conflict resolution
- Designing responsive, accessible UIs with CSS variables and dark mode support
- End-to-end OCR pipeline: image preprocessing → text extraction → LLM parsing

Conclusion

- Successfully built an end-to-end AI-powered receipt digitizer with OCR, LLM extraction and analytics
- Delivered multilingual support, dark mode, admin panel and a real-time AI chatbot
- Implemented secure user authentication with bcrypt, Google OAuth and password reset flow
- Collaborated across 6 members using Git, resolving merge conflicts and integrating features cleanly
- Gained hands-on experience with PaddleOCR, Mistral, Flask and full-stack web development

Thank you!

Team D